

3D Porosity Modeling of a Carbonate Reservoir Using Continuous Multiple-Point Statistics Simulation

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Summary

Training images are numerical representations of geological conceptual models that provide prior information on reservoir architecture. A new emerging geostatistical approach named multiple-point statistics (MPS) simulation allows extracting multiple-point structures from such training images and anchoring these structures to the data actually observed in the reservoir. By reproducing multiple-point statistics inferred from training images, MPS enables the modeling of complex curvilinear structures (e.g., sinuous channels) in a much more geologically realistic way than traditional two-point statistics (variogram-based) techniques. However, in the original MPS implementation, all MPS moments computed from the training image are exported to the reservoir model without processing, which allows simulating only categorical or discretized variables. This implementation is appropriate with clastic reservoirs for which, typically, depositional facies are simulated first using MPS, then porosity and permeability are distributed within each simulated facies using traditional variogram-based techniques. But for reservoirs, in particular in carbonate environments, where porosity and permeability trends/cycles are not closely tied to any facies distribution, simulating porosity/permeability directly using corresponding continuous training images appears to be a more suitable approach.

In this paper, a new filter-based implementation of MPS, named *filtersim*, is proposed to reproduce features from continuous variable training images. First, a set of general filters is applied to the training image to transform (summarize) each training pattern into a set of scores accounting for different aspects of the pattern, such as north-south and east-west gradients and curvatures. The training patterns are classified in the score space and grouped into a small number of similarity classes. The simulation consists then of visiting each grid node along a random path, identifying the similarity class that best fits to the local conditioning data, and patching a pattern drawn from that selected similarity class. In our study, this new approach was applied to model the 3D porosity distribution of a carbonate reservoir in Kazakhstan. First, the original "categorical" MPS program *snestm* was used to model the two main reservoir regions, platform and slope, where the spatial porosity distribution is characterized by significantly different features. Interpreted well markers and seismic data were used to condition the modeling of these two regions. Then porosity was distributed in the platform region using the "continuous" filter-based MPS program *filtersim*, as described previously. The 3D training images used in that second step displayed porosity trends/cycles controlled by the type of geological sedimentation process believed to have occurred in the reservoir.

Introduction

Reservoir modeling aims at depicting spatial heterogeneity distribution of reservoir facies architecture and petrophysical properties such as porosity and permeability based on collected data, which could include core measurements, well-log interpretation, and seismic data. Reservoir static models have to be built numerically such

that reservoir dynamic behavior can be predicted by the aid of reservoir simulations. In order to reflect the hierarchy of reservoir heterogeneities, a two-step modeling approach is typically conducted: Build the facies model first, and then populate petrophysical properties within each facies.

This approach works well for reservoirs the petrophysical properties of which are mainly governed by depositional facies. For example, in clastic reservoirs with fluvial facies distribution, predicting channel locations as well as sizes is more important than interpolating petrophysical properties within channels because their variations are controlled by channel architectures and are therefore considered secondary. Following this principle, most reservoir modeling research focuses on facies modeling, trying to find more efficient methods to delineate the spatial architecture distribution of complex facies association or patterns. Two approaches are typically used to perform stochastic reservoir facies modeling: object-based and pixel-based. Object-based modeling is the most straightforward way to reproduce crisp geometries by parametrizing shape parameters (Haldorsen and Damsleth 1990; Holden et al. 1998), but the major limitation of this approach is that it becomes difficult to condition to local data, particularly if the data are dense with regard the average object size.

In order to reproduce curvilinear facies structures as well as to keep the flexibility of data conditioning, the MPS approach was proposed (Guadiano and Srivastava 1993). MPS allows capturing facies structures from a training image and anchoring them to specific reservoirs. The training images are numerical prior geological models that contain the facies structures and relationships believed to exist in realistic reservoirs. They can be hand-drawn maps or created by computer tools. Training images supply modelers with intuitive or visual facies patterns as an input to facies modeling such that modelers can decide to use them or to generate alternatives. Because MPS is a pixel-based approach, it is easy to condition training patterns to local data.

Once facies models are obtained, petrophysical properties can be populated within each facies. Traditionally, Gaussian simulation is used to predict petrophysical properties. The major limitation of Gaussian simulation is that it only reproduces "maximum entropy" patterns or "maximum randomness structures" (Journel and Deutsch 1993). As a result, low entropy features cannot be reflected in the final petrophysical property models even though a two-step modeling approach is used. The following are three possible reasons that would prevent us from using a two-step modeling approach:

- There is no facies interpretation available for this case study.
- Petrophysical properties are not closely tied to facies (e.g., porosity modification because of diagenesis in carbonate reservoirs).
- Petrophysical property trends/cycles exist and should be reproduced in the models.

In this paper, we propose to use petrophysical property training images, which are continuous variables, as prior models to capture petrophysical property features. Geology analogs and interpretations are used to build these training images.

We apply our approach to model 3D porosity distribution of a carbonate reservoir. According to the geological interpretation, this reservoir is segmented into two gross regions: platform and slope. Seismic data are used to identify the region boundaries. We have some core porosity samples and 15 log porosity curves, but there is no facies interpretation available. From the geological

interpretation and analogs, geologists believe that within the platform region, porosity distribution has cycles/trends, which are governed by depositional processes. In slope region, traditional two-point statistics (variogram-based approach) can be applied to model porosity because of the lack of directional trends. This paper focuses on modeling 3D porosity in the platform region of the carbonate reservoir. The following tasks are critical in this case study:

1. Quantify the prior geological interpretation (i.e., porosity cycles/trends through a porosity training image).

2. Capture the porosity patterns, and anchor them into the reservoir under study with conditioning to the well data. This is performed by applying a set of general filters to the porosity training image to perform training pattern classification and then pattern simulation.

In this paper, we will discuss the procedures used to generate a 3D porosity training image and the process to perform training pattern classification and pattern simulation by filters. Some practical issues are also discussed.

3D Porosity Training Image

Generating a 3D porosity training image is a critical step in our MPS simulation. It allows geologists to incorporate their interpretations into the actual reservoir model. In the carbonate reservoir under study, the following information was available: cycle thickness distribution, core-porosity-sample histograms, and porosity trends within each cycle. The dimension of the porosity training image built is 90×72×196 (1,270,080 pixels), which is of the same size as the reservoir model grid. In MPS simulation, the size of the training image could be different from that of the simulation grid, but the resolution must be the same. In other words, one pixel in the training image must be representative of one pixel in the simulation grid. To improve pattern reproduction in MPS simulation, a large training image that contains more representative or rich patterns is preferred, but this results in more RAM cost and CPU time. Therefore, the generated training image is required to reflect the pattern complexity in the reservoir under study, but should be kept to a reasonable size.

Along the z-direction of the training image grid, the different porosity-cycle blocks are drawn from the thickness interpreted by the geologists. For each such block, two 2D nonconditional simulated realizations of porosity were run, one for the top surface, and another independently for the bottom surface. In order to ensure the porosity trend with low porosity at the bottom and high porosity at the top of each cycle, we ran sequential Gaussian simulations (Deutsch and Journel 1998) (*sGsim*) with a known mean value assigned to the internal simple kriging. The lower quartile value of the porosity distribution from core samples is taken as the mean value for the bottom surface; the upper quartile value of the core samples is used for the top surface. Conditional to these 2D simulated porosity realizations at the top and bottom of each block, a 3D *sGsim* conditional simulation was performed next. Each of these 3D conditional simulations identifies the corresponding porosity histogram obtained from core samples.

Last, by merging all these 3D simulated porosity blocks together, a 3D porosity training image with clear cycles/trends is created. **Fig. 1** shows the 3D porosity training image using *GOCAD* geostatistical software. The porosity cycles/trends can be clearly observed. The range of porosity values from core samples is between 0.0 and 0.25. Other approaches could have been used to generate such training image, as long as the final result does reflect the required cyclicity and porosity statistics.

Pattern Filtering and Classification

Once a 3D porosity training image is prepared, we need to condition these features to well data. We use filters to summarize patterns and distinguish pattern classes. A detailed discussion of filter approach can be found in the paper of Zhang et al. (2006).

Filter Construction. A filter is defined over a template with a set of weights attached to each node. Applying a filter to a local pattern, an averaged value, which is named score, is obtained. Each score is considered a summary of a pattern, resulting in a

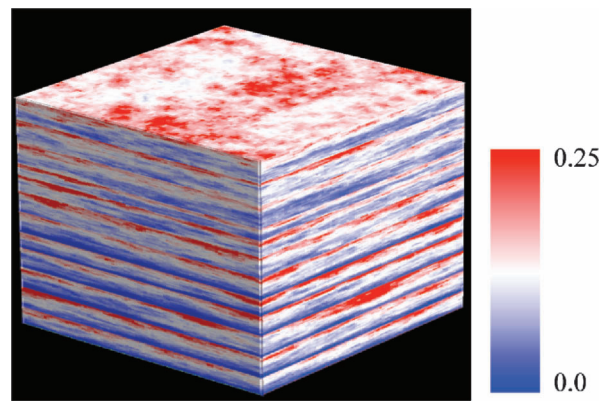


Fig. 1—3D porosity training images for platform.

tremendous dimension reduction. Using a single filter is not enough to distinguish between different training patterns. We proposed nine general filters in 3D (six in 2D). Each filter sees different aspects of the local training pattern. The following is a filter score definition by applying a filter to a local pattern:

$$S_f(i, j, k) = \sum_{w=-nz}^{nz} \sum_{v=-ny}^{ny} \sum_{u=-nx}^{nx} X(i+u, j+v, k+w) f(u, v, w), \dots (1)$$

where X is a local training pattern centered at (i, j, k) ; f is a filter defined over a template with the size of $(2nx+1) \times (2ny+1) \times (2nz+1)$; and S_f is the filter score of the pattern assigned to the center of this template.

We define nine different filters $f_1, \dots, f_9$ as follows.

• f_1 : N-S (north-south) average:

$$f_1(u, v, w) = 1 - \frac{|v|}{n}, v = -n, \dots, n \dots (2)$$

• f_2 : E-W (east-west) average

$$f_2(u, v, w) = 1 - \frac{|u|}{n} \dots (3)$$

• f_3 : T-B (top-bottom) average

$$f_3(u, v, w) = 1 - \frac{|w|}{n} \dots (4)$$

• f_4 : N-S gradient

$$f_4(u, v, w) = \frac{v}{n} \dots (5)$$

• f_5 : E-W gradient

$$f_5(u, v, w) = \frac{u}{n} \dots (6)$$

• f_6 : T-B gradient

$$f_6(u, v, w) = \frac{w}{n} \dots (7)$$

• f_7 : N-S curvature

$$f_7(u, v, w) = \frac{2|v|}{n} - 1 \dots (8)$$

• f_8 : E-W curvature

$$f_8(u, v, w) = \frac{2|u|}{n} - 1 \dots (9)$$

• f_9 : T-B curvature

$$f_9(u, v, w) = \frac{2|w|}{n} - 1 \dots (10)$$

Average filters highlight the pattern locations; gradient filters (order 1 difference) give edge detection or boundary contrast; curva-

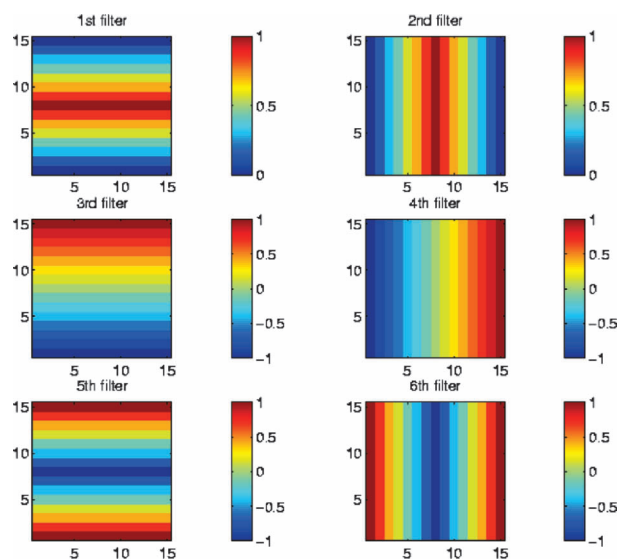


Fig. 2—Six general 2D filters (15×15)

ture filters reflect order 2 transitions. Each set of filters is applied to three directions: North-South, East-West, and Top-Bottom.

Fig. 2 shows the six 2D general filters over a 15×15 template, and each color bar indicates the filter weights.

Pattern Classification. We apply nine filters defined over a 3D template with the size of 9×9×7 one by one to scan the 3D porosity training image of Fig. 1. At each location, a local training pattern is transferred into nine score values. Finally, a nine-dimensional score space is created, and each point in score space corresponds to at least one local porosity training pattern.

In order to classify training patterns, we partition the score space into cells. Each individual score axis is divided into three equal frequency bins, which leads to a total of $3^9 = 19,683$ cells created in the score space. Similar patterns will fall in the same cell because their scores are close.

After score space partitioning, all similar patterns within the same cell are averaged by pixel wise to generate a training-pattern prototype that is considered the center of this pattern class. This allows for fast search for the closest pattern class to a data event in the simulation stage.

Conditional Porosity Simulation

In order to simulate the porosity in the platform region of the reservoir, two regions (platform and slope) were simulated first using MPS algorithm *snesim* developed by Strebelle (2002). *Snesim* algorithm works only with categorical variable training images. Fig. 3 shows the 15 wells with marked platform (red) and slope (white) indicators, as interpreted from well data; Fig. 4 shows one conditional *snesim* simulation of the platform region.



Fig. 4—Simulated platform region using *snesim*.

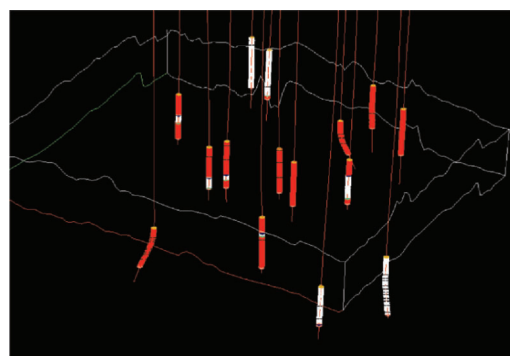


Fig. 3—Wells with platform (red) and slope (white) indicators.

Next, we simulate porosity in the previously simulated platform region conditioning to the 15 well-log porosity data. This simulation step is based on the porosity training pattern classification obtained from filter scores. Sequential simulation approach is performed with each simulation node being visited randomly. At each node, neighborhood conditioning data are searched and compared with each training prototype to pick the closest prototype according to a specified distance function. Once the closest prototype is found, a training pattern from that prototype is drawn and patched to the simulation node without replacing simulated data values. Then the inner part of the patch is frozen and considered as hard data, and it will not be revisited. This procedure proceeds until all nodes are simulated. In our case study, the size of the inner patch is 5×5×3. This is different from the traditional simulation approach, which simulates one single node at a time. Freezing an inner patch makes simulation faster, but it may produce discontinuity if the patched area is too large.

It should be pointed out that multigrids are used to perform the simulation such that different scale features can be captured [see Zhang et al. (2006) for detailed discussion]. We used three levels of multigrids in this case study.

Multigrid simulation is performed with corresponding multi-scale data templates. The template over coarse scale is obtained by rescaling the base template by a factor of 2 along each direction, making the template physically large but keeping the same number of template nodes. A larger template allows capturing of large-scale continuities. All training patterns are classified by filter scores at each one of the multiple grids, and then sequential simulation proceeds over that multiple grid. This process is repeated from coarse to fine grid, passing the simulated values from coarse grid to the next finer grid as conditioning data; however, these coarse-grid values are resimulated at each finer grid. Resimulating those values previously simulated over the coarser grids allows correcting of some mismatched patterns.

One conditional realization is displayed in Fig. 5, and a zoomed-in part is shown in Fig. 6. They are shown in a stratigraphic grid. It is seen that the porosity cycles/trends in the training image are well reproduced.

Practical Issues Related to Data Conditioning

When we checked the data conditioning, some well traces were spotted because of the imperfect match between the simulated porosity in the neighborhood of the wells and the well data (well-logging interpretations); see Fig. 7. Even though high/low value training patterns are indeed captured by high/low well data, they may not match locally. This is because of the fact that the training image contains ideal patterns not matched exactly by the real data, especially dense data along the vertical wells. One may ask the question: Why is the data conditioning “perfect” in two-point statistics or variogram-based simulation approach, such as *sGsim*? The reason lies in that this “perfectness” comes from the maximum entropy structures of *sGsim* simulations so that no discontinuity can be detected in the patterns with maximum randomness beyond

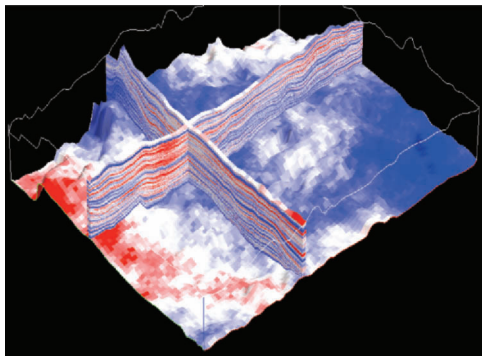


Fig. 5—Simulated porosity in platform region using *filtersim*.

two-point statistics. On the contrary, any mismatch in the low-entropy structure is easily spotted.

To alleviate this mismatch, the conditional *filtersim* porosity realizations were post-processed by running a collocated *sGsim*. The conditional initial realizations are considered as secondary data or soft constraints. The purpose for treating the *filtersim* realization as soft is to keep the porosity cyclicity trend simulated in the realization. The original 15 well-porosity data are considered as hard data. The correlation coefficient between the soft data (the *filtersim* realization) and hard well-logging data was chosen as 0.8 for running the collocated *sGsim* algorithm.

Fig. 8 shows one final realization by post-processing *filtersim* simulation in Fig. 7. It is seen that both the smooth data conditioning and porosity cycles/trends are obtained.

Comparison With Two-Point Statistical Simulations

In order to compare our MPS algorithm *filtersim* with traditional two-point simulation algorithm, a basic *sGsim* is also run to build a 3D porosity model in the platform region. Figs. 9 and 10 are the two enlarged cross sections of two 3D porosity models at the same location from our MPS algorithm and *sGsim*, respectively. It can be seen that MPS realization reproduces clear porosity cycles/trends, while *sGsim* realization only shows a layer-cake porosity distribution.

Conclusions

In this paper, a set of general filters were constructed to classify and simulate patterns from a 3D porosity training image for a carbonate reservoir. The following conclusions can be drawn from this study:

1. MPS allows incorporating of prior geological interpretation into reservoir modeling through training images.
2. In addition to categorical variable training images (e.g., facies), training images can include continuous variables such as porosity, permeability, or any other reservoir properties being modeled. Geological features or textures believed to exist in reser-

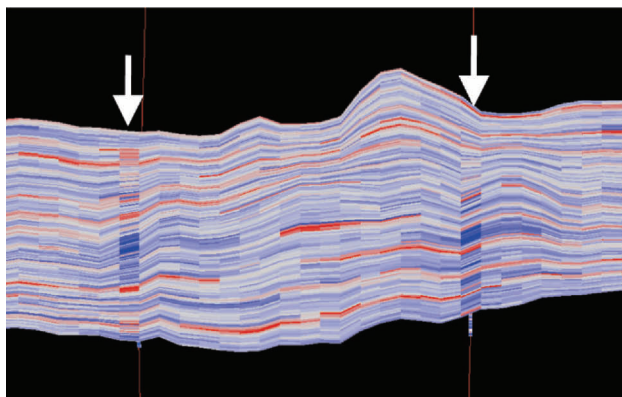


Fig. 7—A check of data conditioning.

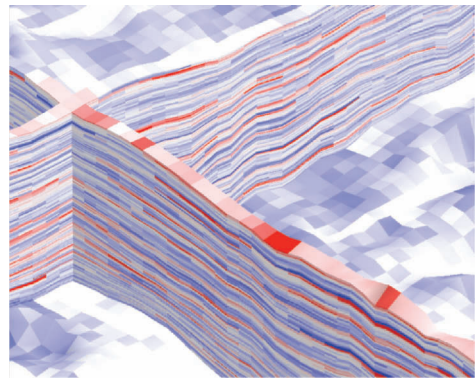


Fig. 6—A closer look at simulation.

voirs can be incorporated into the training images, which are viewed as conceptual geological models.

3. The training patterns can be captured and anchored to local data by our *filtersim* algorithm. It consists of training-pattern classifications and pattern simulations. Our proposed general filters can be applied to any training image.
4. Training images typically contain “idealized” patterns that may not match the data pattern when dense data are available, especially in the case of continuous variables. In this situation, training-image patterns can be considered soft information to constrain the models so that a tradeoff between training patterns and seamless well-data conditioning is achieved.

Nomenclature

- f = filters
- (i, j, k) = center coordinates of filter template
- nx, ny, nz = half size of the filter template along x, y, z direction
- S_f = score for filter f
- (u, v, w) = node coordinates of filter template
- X = local training pattern

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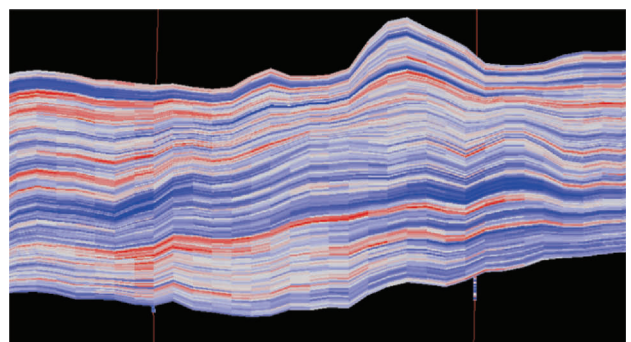


Fig. 8—Porosity simulation after post-processing.

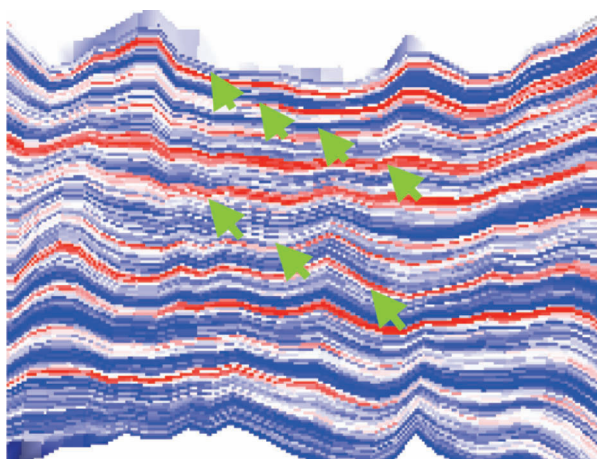


Fig. 9—Porosity simulation reflecting cycles using *filtersim*.

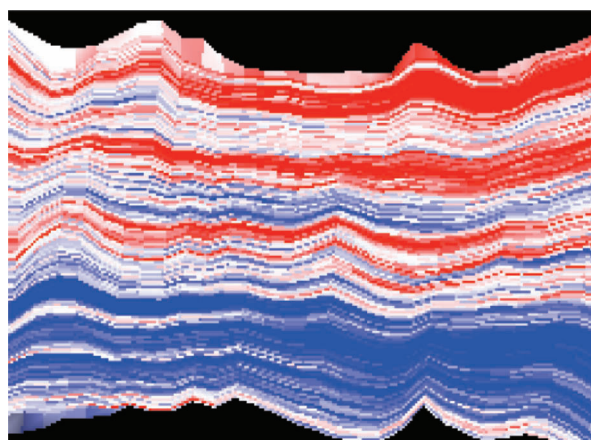


Fig. 10—Porosity simulation from *sGsim*.

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